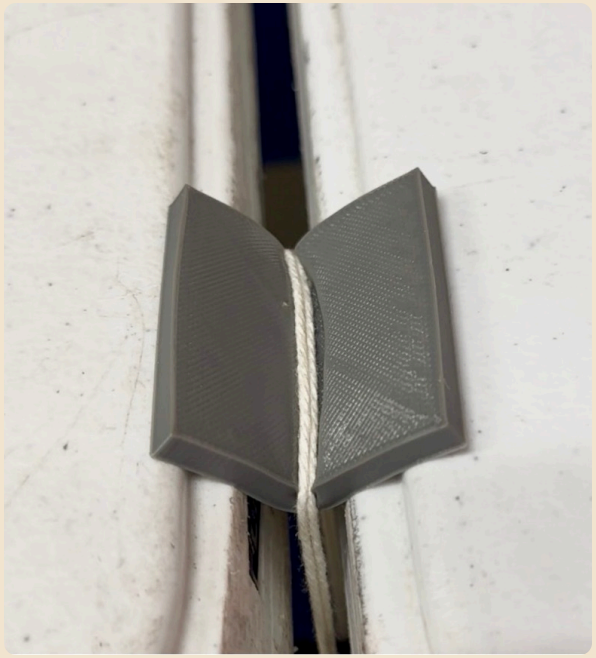
VALUE STATEMENT

WHAT ORTHOPRINT PROVIDES

OrthoPrint gives an injured astronaut a splint made for their own ankle, printed on the lunar base when it is needed. The crew gets real support without launching a box of splints in every size.

Support	Holds the ankle still so a torn ligament can heal.
A true fit	Made from a scan of the patient's own leg.
Independence	Straps the astronaut can tighten with one hand.
Clean wear	Comes off for washing and skin checks.
Less launch mass	Print one splint instead of stocking every size.



Our printed prototype plates during the hanging-weight test

See the full CAD online
gertrudish978-pixel.github.io/lunar-splint

OUR DESIGN · OTHER DESIGN FEATURES

SPECS & MATERIALS

ORTHOPRINT	PROTOTYPE #1 · OCT 2026
INJURY	Lateral ankle ligament tear
FIT	Custom, from a limb scan
CLOSURE	Reusable zip-tie style straps, X pattern
SHELL	Solid wall; small vents being re-tested
PRINTER	Habitat FDM or SLA
SIZE	460.59 mm tall; plates 144 mm apart
MASS	To be measured from the final CAD
PRINT TIME	To be measured from the final CAD

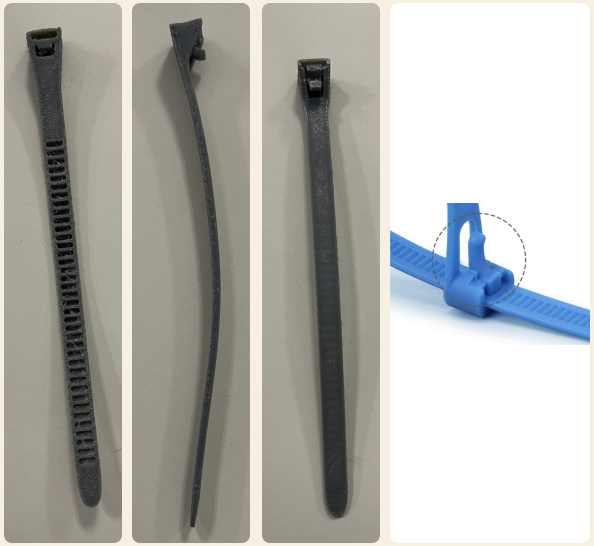
PLA
Prototype 1

ABS
Prototype 2

TPU
Prototype 3 · flexes, firm when solid

PETG
Tested · broke when bent

CF-PEEK
The material we hope to use for flight



Strap, front · Side · Back · Reusable zip tie idea

NEXT STEPS

→ Test strap thickness and the load at which straps break
→ Find the temperature where the filament softens
→ Prototype #2 for the Critical Design Review

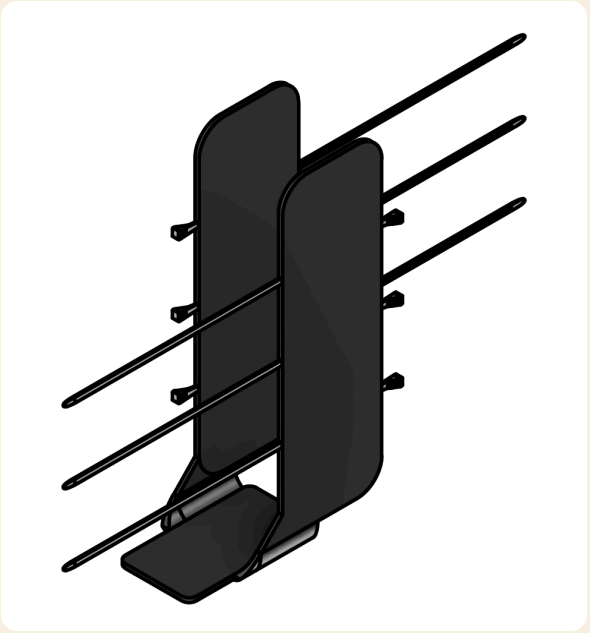
A NASA HUNCH 2026–27 student design project. Not a medical device.

GLADYS PORTER EARLY COLLEGE HIGH SCHOOL
BROWNSVILLE, TX · NASA HUNCH 2026–27

NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

ORTHOPRINT

PDR Preliminary Design Review · Oct 21–22, 2026



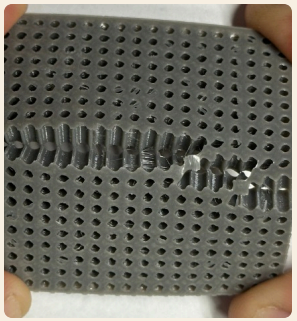
NASA PROJECT	Lunar Splint Design Challenge
PROGRAM	Design & Prototype
SCHOOL	Gladys Porter Early College High School
LOCATION	Brownsville, TX
TEACHERS	Lupita C. Vasquez and Claudia Cortez
TEAM LUNOVA	Gertrudis, Jose and Nancy

Website, videos and CAD
Scan for our testing and prototype videos, CAD and engineering notebook
gertrudish978-pixel.github.io/lunar-splint

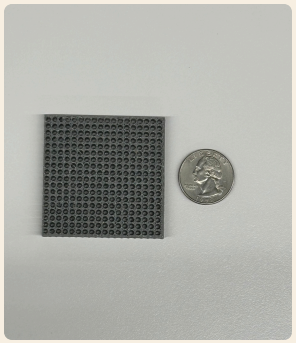
PROTOTYPE HIGHLIGHTS

FROM SCAN TO SUPPORT

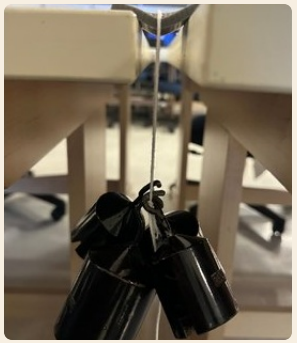
1. **Scan**
   Scan the injured astronaut's lower leg.
2. **Print**
   Print the custom shell and straps on the habitat printer.
3. **Fit**
   Unlatch the splint and fit it on vertically.
4. **Secure**
   Wrap the straps in an X, three times around the ankle.



Sample with holes split, so the wall went solid



TPU sample, quarter for scale



Hanging weights on a sample



Solid TPU, the wall we chose

TEST RESULTS

PETG, holes: broke after bending	REJECTED
TPU, holes: layers split	REJECTED
Solid TPU: firm, moulds slightly	CHOSEN
Straps at key points: ankle stayed still	CHOSEN

OUR DESIGN · UNIQUE ASPECTS

BUILT FOR THE MOON

INNOVATION 1

SINGLE-PIECE RATCHET CLOSURE

Provides a secure hold the astronaut can tighten or release alone, with no loose parts to float away.

INNOVATION 2

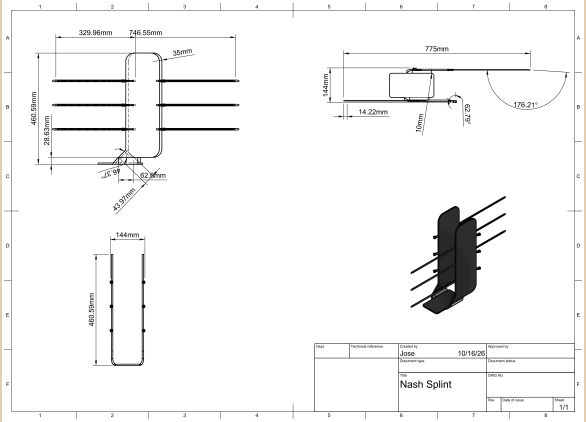
INTEGRATED HYGIENE

Provides a splint that stays clean and lets the skin be checked over weeks of wear.

INNOVATION 3

THERMAL PHASE DESIGN

Provides a shell flexible enough to put on and firm enough to hold the ankle.



Detail drawing, dimensions in mm · Jose, 10/16/2026

OUR DESIGN · BEYOND THE REQUIREMENTS

MORE THAN THE CHALLENGE ASKED FOR

1. **Three lunar innovations**
   The challenge asks for at least one.
2. **Custom fit from a scan**
   Each patient gets their own splint instead of the closest fixed size.
3. **One-handed use**
   The injured astronaut can tighten the straps alone.
4. **Backed by user research**
   A Needs Assessment Survey of 56 students and an interview with our school's athletic trainer.
5. **Four materials tested**
   PLA, ABS, TPU and PETG, with CF-PEEK recommended for flight.

NEEDS ASSESSMENT SURVEY · 56 STUDENTS

Ankle injured most	63%
Can't find a good fit	54%
On their feet a lot	96%
Devices help healing	98%

WHY THE MOON NEEDS IT

A return to Earth takes days, so the crew needs a splint they can make on site.

$1M launch cost per kilogram	**1/6 g** lunar gravity	**−1–2%** bone density per month